

Complete Formula & Understanding Guide

Full first-semester syllabus — every topic explained with intuition. Understand the relationship; the formula follows naturally.

★ **Practice Test** — Topics highlighted with a star badge appeared directly in the 50-question practice test. Pay extra attention to these.

Ch 1 · Quantities & Dimensions

Ch 2 · Vectors

Ch 3 · Kinematics

Ch 4 · Work & Energy

Ch 5 · Rotational Dynamics

Ch 6 · Gravitation

Ch 7 · Circular Motion & SHM

Newton's Laws & Friction

01

Quantities, Units & Dimensions

The language and measurement system of physics

PHYSICAL QUANTITIES — SCALARS VS VECTORS

Theoretical: A *scalar* is fully described by magnitude alone (e.g. temperature, mass, speed, energy, distance). A *vector* requires both magnitude AND direction (e.g. velocity, force, displacement, acceleration). Scalars obey ordinary algebra; vectors obey vector algebra.

- ▶ **Scalar examples:** mass, speed, energy, time, temperature, distance, power, pressure
- ▶ **Vector examples:** velocity, force, displacement, acceleration, momentum, torque, electric field
- ▶ Speed is a scalar; velocity is a vector — same magnitude, but velocity specifies direction
- ▶ Distance is scalar; displacement is vector — displacement is the shortest straight-line path from start to end

FUNDAMENTAL & DERIVED QUANTITIES

Intuition: Physics chooses 7 "base" quantities as building blocks. Every other quantity is built from them by multiplication or division.

QUANTITY	SYMBOL	SI UNIT	ABBREV.
Length	L	metre	m
Mass	M	kilogram	kg
Time	T	second	s
Temperature	θ	kelvin	K
Electric current	I	ampere	A
Luminous intensity	J	candela	cd
Amount of substance	N	mole	mol

- **Derived example:** Force = mass $\times$ acceleration $\rightarrow$ kg·m/s² $\rightarrow$ given the name Newton (N)
- A derived unit that gets a special name: N, J, W, Pa, Hz

DIMENSIONAL ANALYSIS

Intuition: Dimensions are like the "type" of a quantity, independent of the number. You cannot add metres to seconds — types must match. Every valid equation must be dimensionally homogeneous (same dimension on both sides).

QUANTITY	DIMENSION
Velocity	LT ⁻¹
Acceleration	LT ⁻²
Force (F = ma)	MLT ⁻²
Work / Energy	ML ² T ⁻²
Power (W/t)	ML ² T ⁻³
Momentum (mv)	MLT ⁻¹
Pressure (F/A)	ML ⁻¹ T ⁻²
Density (m/V)	ML ⁻³
Angle / Efficiency	dimensionless

3 Uses of Dimensional Analysis:

- ① Check if a formula is correct — both sides must match dimensions.
- ② Derive the form of a formula (up to a dimensionless constant).
- ③ Convert between unit systems.

Example check: Is $v^2 = u^2 + 2as$ dimensionally correct?

$$\text{LHS: } (LT^{-1})^2 = L^2T^{-2}$$

$$\text{RHS: } (LT^{-1})^2 + (LT^{-2})(L) = L^2T^{-2} + L^2T^{-2} \checkmark$$

- Dimensionless constants (like 2, π , $\frac{1}{2}$) have no dimensions — ignore them in checks
- Terms being added/subtracted must have the same dimension

02

Vectors

Quantities with both magnitude and direction — and how to operate on them

REPRESENTATION & COMPONENTS

REPRESENTING A VECTOR

RESOLVING A VECTOR INTO COMPONENTS

★ Q13

A vector in 3D space is written as a combination of three unit vectors: $\hat{i}$ (x-direction), $\hat{j}$ (y-direction), $\hat{k}$ (z-direction). Each is a vector of magnitude exactly 1 pointing along its axis.

General vector A

$$A = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$$

Magnitude (3D Pythagoras)

$$|A| = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

Unit vector in direction of A

$$\hat{A} = A / |A|$$

Why divide by $|A|$? Dividing a vector by its own magnitude strips the size, leaving only the direction as a vector of magnitude 1.

Intuition: Any vector at an angle can be split into two perpendicular components. This is like asking "how much of this vector points East, and how much points North?" Once you have components, all vector operations become simple arithmetic.

Vector of magnitude $|A|$ at angle θ to the x-axis

$$A_x = |A| \cdot \cos\theta \text{ (horizontal component)}$$

$$A_y = |A| \cdot \sin\theta \text{ (vertical component)}$$

Reverse: given components, find angle

$$\theta = \tan^{-1}(A_y / A_x)$$

Exam shortcut: "A vector with component 10m in +x, 10m in +y, 5m in +z" $\rightarrow A = 10\hat{i} + 10\hat{j} + 5\hat{k}$. Magnitude $= \sqrt{(100+100+25)} = \sqrt{225} = 15$ units.

VECTOR ADDITION & SUBTRACTION

ADDING & SUBTRACTING VECTORS

Add component by component

$$A + B = (a_x+b_x)\hat{i} + (a_y+b_y)\hat{j} + (a_z+b_z)\hat{k}$$

Subtraction

$$A - B = (a_x-b_x)\hat{i} + (a_y-b_y)\hat{j} + (a_z-b_z)\hat{k}$$

Resultant of two vectors at angle θ (geometric)

$$R = \sqrt{A^2 + B^2 + 2AB \cdot \cos\theta}$$

Geometric method (Triangle / Parallelogram rule):

Place vectors tip-to-tail; the resultant goes from the first tail to the last tip. The parallelogram rule: draw both from the same point; the diagonal is the resultant.

- $A + B = B + A$ (commutative) — order doesn't matter for addition
- $(A + B) + C = A + (B + C)$ (associative)

- $A - B = A + (-B)$. Negative of B = same magnitude, opposite direction.
- Two perpendicular vectors ($\theta = 90^\circ$): $|R| = \sqrt{A^2 + B^2}$

MULTIPLICATION OF VECTORS

DOT PRODUCT (SCALAR PRODUCT)

$$A \cdot B = |A||B|\cos\theta$$

Component form

$$A \cdot B = a_x b_x + a_y b_y + a_z b_z$$

Intuition: The dot product measures how much A and B "agree" in direction. If they're parallel, full agreement. If perpendicular, zero agreement. This is why $\text{Work} = F \cdot d \cdot \cos\theta$ — only the force along the motion does work.

- Result is a **scalar** (a number, no direction)
- $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$ (parallel → full overlap)
- $\hat{i} \cdot \hat{j} = \hat{i} \cdot \hat{k} = \hat{j} \cdot \hat{k} = 0$ (perpendicular → no overlap)
- $A \cdot B = B \cdot A$ (order doesn't change the result)
- Use it to find angle θ between two vectors: $\cos\theta = (A \cdot B) / (|A||B|)$

CROSS PRODUCT (VECTOR PRODUCT) ★ Q12

Magnitude

$$|A \times B| = |A||B|\sin\theta$$

Component form — expand the 3x3 determinant

$$A \times B = (a_y b_z - a_z b_y) \hat{i} - (a_x b_z - a_z b_x) \hat{j} + (a_x b_y - a_y b_x) \hat{k}$$

Intuition: The cross product gives a new vector that is perpendicular to BOTH A and B. Its magnitude equals the area of the parallelogram formed by A and B. Direction: right-hand rule — curl fingers from A toward B; thumb points in the result's direction. Used for torque ($r \times F$) and angular momentum.

- Result is a **vector**
- $\hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}$ (cyclic — right-hand rule)
- $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$ (parallel → zero area)
- $A \times B = -(B \times A)$ — reversing order reverses direction!
- If A and B are parallel ($\theta = 0^\circ$ or 180°): $|A \times B| = 0$
- If A and B are perpendicular ($\theta = 90^\circ$): $|A \times B| = |A||B|$ (maximum)

Exam method: Write $\det \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$ and expand across the top row. Carefully track the – sign on the $\hat{j}$ term.

DEFINITIONS — AVERAGE VS INSTANTANEOUS

SPEED, VELOCITY & ACCELERATION DEFINED

Theoretical key: These definitions are exact and often tested. Understand them conceptually, not just as formulas.

Average velocity (vector)

$$\bar{v} = \Delta x / \Delta t = (x_2 - x_1) / (t_2 - t_1)$$

Average speed (scalar)

$$\text{speed} = \text{total distance} / \text{total time}$$

Average acceleration

$$\bar{a} = \Delta v / \Delta t = (v_2 - v_1) / (t_2 - t_1)$$

Instantaneous velocity (limit as $\Delta t \rightarrow 0$)

$$v = dx/dt \text{ (slope of } x\text{-}t \text{ graph at that point)}$$

- Average velocity $\neq$ average speed (unless motion is in a straight line with no reversal)
- An object can have zero average velocity but non-zero average speed (e.g. a round trip)
- Uniform motion: constant velocity $\rightarrow$ zero acceleration
- Uniform acceleration: acceleration is constant (e.g. gravity near Earth's surface)

THE 4 SUVAT EQUATIONS ★ Q14–Q18

Where they come from: Start with $a = \Delta v / \Delta t$.

Rearrange $\rightarrow v = u + at$ (Eq.1). Area under v - t graph = displacement $\rightarrow$ integrate $\rightarrow$ Eq.2. Eliminate t from Eq.1 into Eq.2 $\rightarrow$ Eq.3. Average velocity $\times$ time = displacement $\rightarrow$ Eq.4. They're all the same physics, just different rearrangements.

$$\textcircled{1} \quad v = u + at$$

$$\textcircled{2} \quad s = ut + \frac{1}{2}at^2$$

$$\textcircled{3} \quad v^2 = u^2 + 2as$$

$$\textcircled{4} \quad s = \frac{1}{2}(u + v)t$$

EQUATION	KNOWN VARIABLES	MISSING
$\textcircled{1} \quad v = u + at$	$u, a, t \rightarrow v$	s
$\textcircled{2} \quad s = ut + \frac{1}{2}at^2$	$u, a, t \rightarrow s$	v
$\textcircled{3} \quad v^2 = u^2 + 2as$	$u, v, a \rightarrow s \text{ or } u$	t
$\textcircled{4} \quad s = \frac{1}{2}(u+v)t$	$u, v, t \rightarrow s$	a

Only valid when acceleration is constant. Identify 3 known values, select the equation missing the one you don't need.

MOTION UNDER GRAVITY

FREE FALL & VERTICAL PROJECTION ★ Q15

PROJECTILE MOTION ★ Q16

Key insight: Gravity is just SUVAT with $a = g = 9.8 \text{ m/s}^2$ acting downward. Choose a sign convention (up = positive) and stick with it.

Thrown upward with initial speed u (taking up as +ve)

$$a = -9.8 \text{ m/s}^2$$

Time to reach maximum height ($v = 0$ at top)

$$t_{\text{top}} = u / g$$

Maximum height reached

$$H = u^2 / (2g) \text{ [from } v^2 = u^2 - 2gH, \text{ with } v=0]$$

Total time of flight (returns to same level)

$$T_{\text{total}} = 2u / g$$

Speed on return to starting height

$$v_{\text{return}} = u \text{ (same magnitude, now downward)}$$

- At maximum height: **vertical velocity = 0**
- Motion is symmetric: time up = time down
- Acceleration = g throughout — even at the very top (common misconception: acceleration is NOT zero at max height)
- If dropped from rest: $u = 0$, $s = \frac{1}{2}gt^2$, $v = gt$

The master insight: Horizontal and vertical motions are completely independent — they share only time. Horizontal: constant velocity (no force, no acceleration). Vertical: free fall ($a = -g$). Analyze each axis separately, then link via t .

Resolve initial velocity at angle θ to horizontal

$$u_x = u \cdot \cos\theta \text{ (horizontal)} \mid u_y = u \cdot \sin\theta \text{ (vertical)}$$

Horizontal ($a = 0$, uniform motion)

$$x = u_x \cdot t = u \cdot \cos\theta \cdot t$$

Vertical ($a = -g$, free fall)

$$y = u_y t - \frac{1}{2}gt^2 \mid v_y = u_y - gt \mid v_y^2 = u_y^2 - 2gy$$

Time to max height ($v_y = 0$)

$$t_H = u \cdot \sin\theta / g$$

Maximum height

$$H = (u \cdot \sin\theta)^2 / 2g$$

Range (horizontal distance, lands at same height)

$$R = u^2 \cdot \sin 2\theta / g \rightarrow \text{Maximum range at } \theta = 45^\circ$$

- Horizontal velocity **never changes** (no horizontal force)
- Speed at any point: $v = \sqrt{v_x^2 + v_y^2}$
- Complementary angles give same range: θ and $(90^\circ - \theta)$ have equal range

GRAPHS OF MOTION

DISTANCE-TIME & VELOCITY-TIME GRAPHS ★ Q17

Intuition: The slope (gradient) of a graph is the rate of change of what's on the y-axis with respect to the x-

RELATIVE MOTION

Intuition: Velocity is always measured relative to a reference frame. A car moving at 60 km/h relative to

axis. Area under a graph accumulates the product of both axes.

GRAPH	SLOPE GIVES	AREA GIVES
x-t (position-time)	Velocity	—
v-t (velocity-time)	Acceleration	Displacement
a-t (acceleration-time)	Rate of change of a	Velocity change

- **Straight line on x-t:** constant velocity (uniform motion), slope = v
- **Curve (parabola) on x-t:** uniformly accelerated motion
- **Horizontal line on v-t:** constant velocity, zero acceleration
- **Straight line on v-t:** uniform acceleration, slope = a
- **Object at rest:** horizontal line on x-t (slope = 0)
- Negative slope on v-t = deceleration (slowing down)

Average velocity from graph (Q17 type)

$$v_{\text{avg}} = (x_2 - x_1) / (t_2 - t_1)$$

the ground appears stationary to a passenger in an identical car moving at 60 km/h in the same direction.

Velocity of A relative to B

$$v_{AB} = v_A - v_B$$

- Same direction: relative speed = $v_A - v_B$
- Opposite direction: relative speed = $v_A + v_B$
- Rain problem: v_{rain} relative to person = $v_{\text{rain}} - v_{\text{person}}$ (vector subtraction)

NL

Newton's Laws, Friction & Elevators

The forces that cause and change motion

NEWTON'S THREE LAWS

Theoretical — these appear as definition questions:

1st Law (Inertia): A body remains at rest or in uniform motion in a straight line unless acted upon by an external net force. Inertia is the resistance to change in motion — heavier objects have more inertia.

2nd Law: Net force = rate of change of momentum.
For constant mass: $F = ma$.

FRICTION ★ Q24

Intuition: Friction acts parallel to surfaces in contact, opposing relative motion. The normal force N is the surface's push perpendicular to itself. The ratio of friction to normal force is μ , which depends on the materials.

$$f = \mu N$$

3rd Law: Every action has an equal and opposite reaction. Forces always come in pairs on different bodies.

$$F_{\text{net}} = ma \rightarrow a = F_{\text{net}} / m$$

- F, a are vectors — direction matters. Net force zero → zero acceleration (not zero velocity)
- 3rd law pairs act on DIFFERENT objects — they never cancel each other
- Weight $W = mg$ is the gravitational force on an object (a force, not a mass)

On a horizontal surface ($N = mg$)

$$f = \mu mg \rightarrow \mu = f / (mg)$$

- **Static friction:** prevents motion from starting. Can be anywhere from 0 to $\mu_s N$
- **Kinetic friction:** acts once object is sliding. $f_k = \mu_k N$ (constant)
- $\mu_s > \mu_k$ always — harder to start sliding than to keep it sliding
- "Just sufficient to move" → applied force = maximum static friction = μN
- On incline: $N = mg \cos \theta$, friction = $\mu mg \cos \theta$, component of gravity along slope = $mg \sin \theta$

ELEVATOR / LIFT PROBLEMS ★ Q25

Intuition: In an elevator, your apparent weight (what a scale reads) changes because you're accelerating. The scale measures the normal force N , not your actual weight mg .

Accelerating upward (a upward)

$$N = m(g + a) \rightarrow N > mg \text{ (feels heavier)}$$

Accelerating downward (a downward)

$$N = m(g - a) \rightarrow N < mg \text{ (feels lighter)}$$

Free fall ($a = g$ downward)

$$N = 0 \rightarrow \text{weightlessness}$$

Motor force for full elevator system

$$F_{\text{motor}} = M_{\text{total}} \times (g + a) \text{ where} \\ M_{\text{total}} = \text{elevator} + \text{all passengers}$$

BRAKING & CAR PROBLEMS ★ Q18

Constant braking force F on car of mass m

$$a = -F/m \text{ (deceleration)}$$

Find mass: car at speed u stops in time t

$$v = 0: 0 = u + at \rightarrow a = -u/t \rightarrow m = \\ F/|a| = Ft/u$$

- Always take deceleration as negative when using SUVAT
- If force opposes motion: net a is opposite to velocity → object slows

WORK DONE ★ Q26 Q33 Q35 Q36

Intuition: Work is only done by the component of force that acts IN the direction of motion. Carrying a bag horizontally does zero work against gravity — weight acts downward, motion is horizontal (90° apart $\rightarrow \cos 90^\circ = 0$).

$$W = F \cdot d \cdot \cos\theta$$

$\theta = 0^\circ$: force parallel to motion (maximum work)

$$W = Fd$$

$\theta = 90^\circ$: force perpendicular to motion

$W = 0$ (e.g. gravity while moving horizontally)

$\theta = 180^\circ$: force opposes motion

$W = -Fd$ (negative work, e.g. friction)

Inclined plane: Work against gravity = mgh where h = VERTICAL height only. NOT the slant length. The slant length is longer, but gravity only cares about vertical rise.

Chain pulled to table — hanging length L_h , linear density m/L

$$W = (m/L) \cdot L_h \cdot g \cdot (L_h / 2) = m_{\text{hang}} \cdot g \cdot (L_h/2)$$

Chain trick: The CM of the hanging part is at its midpoint — $L_h/2$ below the table. You raise it to the table surface, so it rises by $L_h/2$. $W = m_{\text{hang}} \times g \times \text{rise}$.

POWER & EFFICIENCY ★ Q34 Q37

Power = rate of doing work

$$P = W / t = F \cdot v$$

Why $P = Fv$? $P = W/t = (F \cdot d)/t = F \cdot (d/t) = F \cdot v$. Just substitute $v = d/t$.

KINETIC & POTENTIAL ENERGY ★ Q27

Kinetic Energy

$$KE = \frac{1}{2}mv^2$$

Derivation: $F = ma$, $W = Fs = mas$. Using $v^2 = u^2 + 2as \rightarrow s = v^2/2a$. So $W = ma \times v^2/2a = \frac{1}{2}mv^2$. Work done = KE gained. ✓

Gravitational Potential Energy

$$PE = mgh$$

Work–Energy Theorem

$$W_{\text{net}} = \Delta KE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

Conservation (no friction/air resistance)

$$KE_1 + PE_1 = KE_2 + PE_2$$

- At highest point of free flight: all energy is PE, zero KE
- At lowest point: all energy is KE (if starting from rest at height h)
- Friction converts mechanical energy to heat — total energy still conserved, just some becomes thermal

MOMENTUM & COLLISIONS ★ Q27 Q41

Momentum (vector)

$$p = mv$$

Newton's 2nd Law (more general form)

$$F = \Delta p / \Delta t = d(mv)/dt$$

Lifting a mass m at constant speed v

$$P = mgv \text{ (force = } mg, \text{ speed = } v)$$

Efficiency

$$\eta = (\text{Useful output energy} / \text{Total input energy}) \times 100\%$$

Energy wasted as heat

$$W_{\text{heat}} = W_{\text{input}} \times (1 - \eta/100)$$

- Single-string pulley: $MA = 1$, effort = load. $P = \text{load} \times \text{speed}$
- 60% efficient $\rightarrow$ 40% wasted: 100 J in $\rightarrow$ 60 J useful, 40 J heat
- Units: Power $\rightarrow$ Watt (W) = J/s = $\text{kg}\cdot\text{m}^2\cdot\text{s}^{-3}$

Conservation of momentum (no external forces)

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Perfectly inelastic (stick together)

$$m_1u_1 + m_2u_2 = (m_1 + m_2)v_f$$

Impulse-momentum theorem

$$J = F\Delta t = \Delta p = mv - mu$$

Elastic vs Inelastic: Both types conserve momentum.

Elastic collisions also conserve KE. *Inelastic* collisions lose KE to deformation, heat, sound. In a perfectly inelastic collision, the objects stick together — maximum KE loss.

- Equal-mass elastic: balls swap velocities (ball 1 stops, ball 2 moves at u_1)
- Momentum is conserved even in explosions (rocket propulsion, gun recoil)
- Momentum of A before collision = $m_A \times u_A$ (independent of B)

05

Rotational Dynamics of Rigid Bodies

Every linear concept has a rotational twin — learn one, get the other free

The Core Parallel: Replace mass with moment of inertia I , force with torque τ , velocity with ω , and displacement with θ (in radians). Every linear equation becomes a rotational one.

LINEAR ↔ ROTATIONAL — MASTER EQUIVALENCE TABLE ★ Q38–Q44

LINEAR QUANTITY	FORMULA	ROTATIONAL EQUIVALENT	FORMULA	LINK BETWEEN THEM
Displacement s	metres	Angular displacement θ	radians	$s = r\theta$
Velocity v	m/s	Angular velocity ω	rad/s	$v = r\omega$
Acceleration a	m/s ²	Angular acceleration α	rad/s ²	$a = r\alpha$
Mass m	kg	Moment of Inertia I	$\text{kg}\cdot\text{m}^2$	$I = \sum mr^2$
Force $F = ma$	N	Torque $\tau = I\alpha$	N·m	$\tau = rF\sin\theta$

Momentum $p = mv$	kg·m/s	Angular momentum $L = I\omega$	kg·m ² /s	$L = mvr$ (point)
KE = $\frac{1}{2}mv^2$	J	Rot. KE = $\frac{1}{2}I\omega^2$	J	—
Work $W = Fs$	J	Work $W = \tau\theta$	J	θ in radians!
Power $P = Fv$	W	Power $P = \tau\omega$	W	—

MOMENT OF INERTIA I
★ Q38 Q43

Intuition: I is rotational inertia — resistance to spinning. Mass farther from the axis is harder to spin (hence r^2). A figure skater pulls arms in → smaller r → smaller I → faster spin. Same angular momentum, less I → more ω .

General definition
 $I = \sum m_i r_i^2$ (sum of mass × radius² for each particle)

SHAPE / AXIS	I
Point mass at radius r	mr^2
Solid disk / cylinder (central axis)	$\frac{1}{2}MR^2$
Hollow cylinder / ring (central axis)	MR^2
Solid sphere (diameter axis)	$\frac{2}{5}MR^2$
Hollow sphere (diameter axis)	$\frac{2}{3}MR^2$
Thin rod (about centre)	$\frac{1}{12}ML^2$
Thin rod (about one end)	$\frac{1}{3}ML^2$

Q38 type: $I = mr^2$. If mass halved AND radius same → I halved ($\frac{1}{2}$ as much). If radius halved AND mass same → I quartered ($\frac{1}{4}$ as much). The r^2 makes radius changes dramatic.

RADIUS OF GYRATION K

Intuition: Instead of thinking about where every bit of mass is, imagine squashing all the mass to a single ring at radius k . That ring would have the same I as the real object. k is the "effective radius" for rotation.

$I = Mk^2 \rightarrow k = \sqrt{I/M}$

- Solid disk: $k = R/\sqrt{2} \approx 0.707R$
- Solid sphere: $k = R\sqrt{2/5} \approx 0.632R$
- k is always less than the maximum radius of the body

PARALLEL & PERPENDICULAR AXES THEOREMS

Parallel Axes Theorem — shift axis by distance d from CM

TORQUE & ANGULAR KINEMATICS
★ Q39 Q40

Torque (turning force)

$$I = I_{\text{cm}} + Md^2$$

Intuition: Moving the axis away from the CM always increases I — the mass now has to "swing" around a farther point. The extra term Md^2 accounts for this. I_{cm} is the minimum possible I for that object.

Verification: thin rod about end = about centre + Md^2

$$\frac{1}{3}ML^2 = \frac{1}{12}ML^2 + M(L/2)^2 = \frac{1}{12}ML^2 + \frac{1}{3}ML^2 \quad \checkmark$$

Perpendicular Axes Theorem — flat objects only

$$I_z = I_x + I_y \quad (\text{z-axis perpendicular to the flat object})$$

- Perpendicular axes theorem: ONLY valid for planar (2D, flat) objects
- x and y lie in the plane of the object; z points out of it

$$\tau = I\alpha = rF \cdot \sin\theta = r \times F$$

The 3 angular SUVAT equations (identical structure!)

$$\omega = \omega_0 + \alpha t$$

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

Always convert revolutions → radians: 1 rev = 2π rad. So 5 rev = 10π rad. 30 rev/s = 60π rad/s. Equations fail if you use revolutions directly.

- $\alpha = \tau/I$ (rotational Newton's 2nd law)
- Torque is maximum when force is perpendicular to the radius ($\sin\theta = 1$)
- Zero torque when force is directed through the axis ($r = 0$ or $\theta = 0^\circ$)

ANGULAR MOMENTUM & CONSERVATION ★ Q28

Angular momentum

$$L = I\omega$$

For a point mass at radius r

$$L = mvr$$

Conservation (no external torque)

$$I_1\omega_1 = I_2\omega_2$$

Finding ω for a moving body with given L

$$\omega = mvr / I \quad (\text{angular momentum} = I\omega = mvr)$$

Skater analogy: No external torque acts on a spinning skater. Pulling arms in reduces $r \rightarrow$ reduces $I \rightarrow \omega$ increases to keep $L = I\omega$ constant. This is conservation of angular momentum in action.

ROTATIONAL KE, WORK & ROLLING ★ Q44

Rotational Kinetic Energy

$$KE_{\text{rot}} = \frac{1}{2}I\omega^2$$

Work done by a torque (θ must be in radians)

$$W = \tau \cdot \theta$$

Rolling without slipping — link linear and angular

$$v = r\omega \rightarrow \text{Total KE} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

Q28 type — Work done by net torque: $W = \tau \times \theta$. Convert revolutions to radians first. Then $W = \Delta KE_{\text{rot}} = \frac{1}{2}I(\omega_2^2 - \omega_1^2)$.

Universal Gravitation

The force that governs planets, moons, satellites — and your weight

NEWTON'S LAW OF GRAVITATION ★ Q31 Q32

Intuition: Every mass in the universe attracts every other mass. The force doubles if either mass doubles, but quarters if distance doubles (inverse square law — like light spreading over a sphere).

$$F = GMm / r^2$$

G = Universal gravitational constant

$$G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 \cdot \text{kg}^{-2}$$

Gravitational field strength (intensity) at distance r

$$g = F/m = GM / r^2 \text{ (units: N/kg} \equiv \text{m/s}^2\text{)}$$

- g on Earth surface $\approx 9.8 \text{ m/s}^2$ (take $R_E = 6.4 \times 10^6 \text{ m}$)
- g on Moon surface $\approx 1.6 \text{ m/s}^2$
- g decreases as $1/r^2$ as you move away from Earth
- Gravitational field intensity = $g = F/m$ — force per unit mass at a point

KEPLER'S LAWS ★ Q30

These are observational laws (Kepler described them before Newton explained why). They are commonly tested as definitions.

- 1st Law — Orbit Shape:** Planets orbit the Sun in ellipses, with the Sun at one focus (not the centre).
- 2nd Law — Equal Areas:** A line from a planet to the Sun sweeps equal areas in equal time intervals. → Planet moves faster when closer to the Sun, slower when farther.
- 3rd Law — Harmonics:** $T^2 \propto r^3$. The orbital period squared is proportional to the orbital radius cubed.

Kepler's 3rd Law (exact)

$$T^2 = (4\pi^2 / GM) \cdot r^3$$

Comparing two orbits

$$T_1^2 / T_2^2 = r_1^3 / r_2^3$$

ORBITAL MECHANICS & SATELLITES

Intuition: A satellite orbits because it's in continuous free fall — it falls toward Earth but moves sideways fast enough to keep missing it. Centripetal force = gravitational force.

Orbital speed (set $F_{\text{gravity}} = F_{\text{centripetal}}$)

$$GMm/r^2 = mv^2/r \rightarrow v = \sqrt{GM/r}$$

Orbital period

$$T = 2\pi r/v = 2\pi\sqrt{r^3/GM}$$

Geostationary orbit: $T = 24 \text{ hours}$ (satellite stays over one spot)

WEIGHTLESSNESS & ESCAPE VELOCITY

Escape velocity (minimum speed to escape gravity)

$$v_{\text{esc}} = \sqrt{2GM/R} = \sqrt{2gR}$$

Derivation: At escape, total energy = 0 (just barely escapes). $KE = PE$: $\frac{1}{2}mv^2 = GMm/R \rightarrow v = \sqrt{2GM/R}$. For Earth: $v_{\text{esc}} \approx 11.2 \text{ km/s}$.

Weightlessness explained: Astronauts in orbit are NOT outside gravity — they're in continuous free fall around Earth. Gravity accelerates both them and the spacecraft identically. Since neither accelerates relative to the other, apparent weight = 0. True

$r_{\text{geo}} = (GM^2/4\pi^2)^{1/3} \approx 42,300 \text{ km}$
from Earth's centre

- Higher orbit → slower orbital speed but longer period
- Geostationary satellites must orbit above the equator, going eastward

weightlessness would require infinite distance from all masses.

COMPARING GRAVITY ON DIFFERENT PLANETS

★ Q29

Surface gravity from mass and radius

$$g = GM/R^2$$

Ratio to Earth (useful shortcut)

$$g_p / g_E = (M_p/M_E) \times (R_E/R_p)^2$$

Venus example: $M = 0.0558 M_E$, diameter = $0.381 D_E \rightarrow R_p = 0.381 R_E / 2 \times 2 = 0.381 R_E$? No: diameter ratio 0.381 → $R_{\text{ratio}} = 0.381/2 \times 2 = \text{same ratio } 0.381$ (if diameter scales, radius scales equally)

$$g_V = 9.8 \times (0.0558 / 0.381^2) \approx 9.8 \times 0.384 \approx 3.8 \text{ m/s}^2$$

Strategy: Diameter given → halve to get radius ratio. Then use the formula above. Pick the answer option closest to your result.

07

Circular Motion & Simple Harmonic Motion

Repeating, oscillating, orbiting — the physics of cycles

UNIFORM CIRCULAR MOTION

CENTRIPETAL ACCELERATION & FORCE

★ Q22 Q23

Intuition: Even at constant speed, circular motion involves acceleration — because direction changes continuously. This acceleration always points toward

BANKING ANGLE

★ Q22

Intuition: On a banked curve, the road is tilted so the normal force has a horizontal component pointing toward the centre — this provides the centripetal

the centre (centripetal = "centre-seeking"). It's provided by whatever force keeps the object on the circle: tension, gravity, friction, or normal force.

Centripetal acceleration

$$a_c = v^2/r = \omega^2 r$$

Centripetal force

$$F_c = mv^2/r = m\omega^2 r$$

Linking linear speed v , angular velocity ω , period T

$$v = \omega r \mid \omega = 2\pi/T = 2\pi f \mid T = 2\pi/\omega$$

- "Centrifugal force" is not a real force — it's an illusion felt in the rotating frame
- For a stone on a string: Tension $T = mv^2/r =$ centripetal force
- String cuts when $T \geq$ breaking tension: find $v_{\text{max}} = \sqrt{(T_{\text{max}} \cdot r / m)}$
- For a car on a flat curve: friction provides centripetal force

force without needing friction. The bank angle is optimized for one specific speed.

$$\tan\theta = v^2 / (rg)$$

Normal reaction (wheels perpendicular to banked surface)

$$N = mg / \cos\theta \rightarrow \text{total reaction on track}$$

Proof: Resolve N : Vertical component $N \cdot \cos\theta = mg \rightarrow N = mg/\cos\theta$. Horizontal component $N \cdot \sin\theta = mv^2/r$. Divide: $\sin\theta/\cos\theta = v^2/(rg) \rightarrow \tan\theta = v^2/(rg)$.

SIMPLE HARMONIC MOTION (SHM)

WHAT IS SHM? — THEORY ★ Q20 Q21

Definition: SHM is periodic motion where the restoring force (and hence acceleration) is always directed toward a fixed equilibrium point AND is proportional to displacement from it.

Condition: $a = -\omega^2 x$. The minus sign is critical — it means the acceleration always opposes displacement (always pulls back toward centre).

- Examples: mass on spring, simple pendulum (small angles), vibrating molecules, alternating current
- Period T is constant regardless of amplitude (for ideal SHM)

SHM EQUATIONS OF MOTION ★ Q45-Q50

Displacement (starting from centre, $x=0$ at $t=0$)

$$x = A \cdot \sin(\omega t)$$

Displacement (starting from extreme, $x=A$ at $t=0$)

$$x = A \cdot \cos(\omega t)$$

General form with initial phase ϕ

$$x = A \cdot \sin(\omega t + \phi)$$

Velocity at any displacement x

$$v = \omega \sqrt{(A^2 - x^2)}$$

At centre ($x=0$): maximum velocity

$$v_{\text{max}} = \omega A$$

- At equilibrium ($x=0$): maximum velocity, zero acceleration, zero PE, maximum KE
- At extreme ($x=\pm A$): zero velocity, maximum acceleration, maximum PE, zero KE
- Total energy $E = KE + PE = \text{constant}$ throughout the motion

At extreme ($x=\pm A$): maximum acceleration

$$a_{\text{max}} = \omega^2 A$$

Q47 type — equation from given data: Given $A = 0.5$ cm, $T = 0.4$ s, initial phase $= \pi/2$. Compute $\omega = 2\pi/T = 2\pi/0.4 = 5\pi$ rad/s. Equation: $y = 0.5 \cdot \sin(5\pi t + \pi/2) = 0.5 \cdot \cos(5\pi t)$ [since $\sin(x + \pi/2) = \cos x$].

SHM PERIOD, FREQUENCY & ENERGY

★ Q48 Q49 Q50

$$\omega = 2\pi/T = 2\pi f \quad | \quad T = 2\pi/\omega \quad | \quad f = \omega/2\pi$$

Spring-mass system

$$\omega = \sqrt{k/m} \rightarrow T = 2\pi\sqrt{m/k}$$

Simple pendulum (small angles only)

$$T = 2\pi\sqrt{L/g}$$

Total mechanical energy

$$E = \frac{1}{2}kA^2 = \frac{1}{2}m\omega^2 A^2$$

Velocity at the centre (for oscillator with period T and amplitude A)

$$v_{\text{max}} = A\omega = A \cdot (2\pi/T) = 2\pi A/T$$

Q48 type: Period $= 0.01$ s, $A = 0.2$ m $\rightarrow \omega = 2\pi/0.01 = 200\pi$ rad/s $\rightarrow v_{\text{max}} = 0.2 \times 200\pi = 40\pi$ m/s ✓

SHM TIME CALCULATIONS

★ Q45 Q46

How to find time for any position: Use $x = A \cdot \sin(\omega t)$ and solve for t . The standard results below all come from inverting this sine function.

JOURNEY (STARTING FROM CENTRE)	TIME	HOW
Centre $\rightarrow$ Amplitude A (quarter cycle)	$T/4$	$\sin(\omega t) = 1 \rightarrow \omega t = \pi/2 \rightarrow t = T/4$
Centre $\rightarrow$ Half amplitude (A/2)	$T/12$	$\sin(\omega t) = 1/2 \rightarrow \omega t = \pi/6 \rightarrow t = T/12$
Centre $\rightarrow A/\sqrt{2}$	$T/8$	$\sin(\omega t) = 1/\sqrt{2} \rightarrow \omega t = \pi/4 \rightarrow t = T/8$
Full oscillation (one complete cycle)	T	—

Q46 type — find T from two (x, v) pairs using $v^2 = \omega^2(A^2 - x^2)$

$$v_1^2 = \omega^2(A^2 - x_1^2) \quad \text{and} \quad v_2^2 = \omega^2(A^2 - x_2^2)$$

Subtract to eliminate A^2

$$v_1^2 - v_2^2 = \omega^2(x_2^2 - x_1^2) \rightarrow \omega^2 = (v_1^2 - v_2^2)/(x_2^2 - x_1^2) \rightarrow T = 2\pi/\omega$$

SIMPLE PENDULUM & MOON CLOCK

★ Q19 Q30

$$T = 2\pi\sqrt{L/g}$$

What affects the pendulum period: Only L (length) and g (gravity). NOT mass, NOT amplitude (for small angles). This is why Galileo noticed all pendulums of the same length swing at the same rate regardless of what they're made of.

- Longer pendulum → longer period (swings more slowly)
- Stronger gravity → shorter period (swings faster)
- On Moon ($g_M = 1.6 \text{ m/s}^2 < g_E$): period is longer → clock runs slow

Moon clock — how many hours it records per Earth hour

$$t_{\text{Moon}} = (g_M / g_E) \times 1 \text{ h} = (1.6/9.8) \text{ h per Earth hour}$$

Reasoning: $T_{\text{Moon}}/T_{\text{Earth}} = \sqrt{g_E/g_M} \approx 2.47$.

The Moon pendulum takes 2.47× longer per swing. In one Earth hour it completes fewer swings → records fewer hours. It records $T_{\text{Earth}}/T_{\text{Moon}} = \sqrt{g_M/g_E} = \sqrt{1.6/9.8} = 1.6/9.8$ hours.



Quick-Fire Reference — What You Must Know Cold

CONSTANTS, CONVERSIONS & EXAM TRAPS

ESSENTIAL CONSTANTS		DIMENSIONS	
g (Earth)	9.8 m/s ² (or 10 if stated)	Force	MLT ⁻²
g (Moon)	1.6 m/s ²	Energy / Work	ML ² T ⁻²
G	$6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$	Power	ML ² T ⁻³
1 revolution	2π radians	Momentum	MLT ⁻¹
1 km/h	÷ 3.6 = m/s	Pressure	ML ⁻¹ T ⁻²

- **Inclined plane:** $W = mgh$, h = vertical height only
- **Chain:** $W = m_{\text{hang}} \times g \times (L_{\text{hang}}/2)$

- **SHM centre** → $A/2$: time = $T/12$
- **Equal-mass elastic collision**: swap velocities
- **Banking**: $\tan\theta = v^2/rg$; $N = mg/\cos\theta$
- **Parallel axes**: $I = I_{\text{cm}} + Md^2$
- **Rolling**: $v = r\omega$; $KE = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$
- **Moon clock**: records 1.6/9.8 h per Earth hour
- **Max acceleration in SHM**: at extremes ($x = \pm A$)
- **Max velocity in SHM**: at centre ($x = 0$)